Emil Kozicki

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SUMMARY

Backend-focused software engineer with production experience building and supporting AWS-based integrations and data pipelines in Python. Strongest at diagnosing production failures across systems (AWS, Salesforce, third-party vendor APIs) and shipping features end to end, from data model to UI. M.S. in Computer Science from UNC Charlotte.

PROFESSIONAL EXPERIENCE

Bridge Investment Group Charlotte, NC

Software Engineer *February 2026 – September 2026*

- Supported a fleet of AWS Lambda integrations syncing data between Salesforce and third-party property-management vendors: triaged production failures via Jira and CloudWatch, reconciled drift between live Lambda code and source control, added retry logic for dropped Salesforce connections during large paginated queries, and extended vendor CSV parsing to support new fields without breaking existing exports.
- Designed and shipped a Deferred Work Items feature end to end, including the Salesforce data model (Apex changes, a new lifecycle-event object to track item history) and client-side schema migrations and UI in HomeWyze, an existing offline-first SvelteKit PWA.
- Resolved a production incident caused by the Deferred Work Items schema change, where devices running stale cached app code submitted payloads that no longer matched the updated backend schema: patched the Lambda to accept both formats, recovered failed submissions by re-running Step Functions executions, and built a background update-detection service that keeps clients on current code without interrupting users.
- Monitored and maintained AWS Batch job scheduling across integration pipelines. Diagnosed a queue-wide outage where a job definition's memory requirement exceeded every attached compute environment's capacity, silently blocking ~290 queued jobs; restored the last known-good configuration and added an EventBridge/SNS alert on Batch's queue-blocked events to catch this failure mode automatically.
- Designed and shipped a bulk photo-export feature for HomeWorx, a Salesforce-based property inspection platform, spanning a Lightning Web Component, an Apex integration layer with sandbox/production routing and error handling, and a Python Lambda that downloads S3-stored photos concurrently (ThreadPoolExecutor), archives them, and returns a presigned URL for direct download.

PROJECTS

AI Portfolio Site *React*, *Vite*, *Django REST Framework*, *Google Gemini API* (repo)

A React front end backed by a Django REST API, with an AI assistant that answers visitor questions about my background.

- Built a Django REST endpoint that proxies chat requests to the Gemini API, keeping the API key server-side so it never reaches the browser.
- Hardened the assistant against prompt injection and jailbreak attempts by isolating untrusted visitor input in delimited tags, declaring no tools, and enforcing explicit topic and confidentiality boundaries in the system prompt.
- Layered abuse protection over the public endpoints: IP-based burst and daily throttles, request length caps, server-side conversation history truncation, and output token limits.
- Implemented a contact pipeline that persists submissions to the database before sending SMTP notifications, so messages survive email delivery failures.
- Built the front end around CSS custom properties for instant dark/light theming, with canvas-based particle animation and scroll-triggered reveals.

EDUCATION

University of North Carolina at Charlotte May 2024 – August 2025

M.S. in Computer Science

Early Entry Program: began graduate coursework while completing the B.S.

University of North Carolina at Charlotte January 2023 – December 2024

B.S. in Computer Science

Central Piedmont Community College August 2018 – May 2022

Associate of Arts

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, HTML, CSS

Cloud & Backend: AWS Lambda, S3, Step Functions, AWS Batch, Amazon EventBridge, SNS, CloudWatch, AWS CDK, PostgreSQL, MongoDB

Integration/Data: Salesforce, Apex, SOQL, Salesforce Bulk API 2.0, Lightning Web Components (LWC), REST API Integration, ETL/Data Pipelines

Frameworks: Django, Django REST Framework, React, Node.js, Express